

1.1

V ľubovoľnom komplexnm vektorovm priestore V se skalrnm souinm $\langle u, v \rangle$ plat (pro ľubovoľn $u, v \in V$):

$$\operatorname{Re}\langle u, v \rangle = \frac{1}{4} (\|u + v\|^2 - \|u - v\|^2), \quad (1)$$

$$\operatorname{Im}\langle u, v \rangle = \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2). \quad (2)$$

Rieenie

Pre ľubovoľn $z = a + bi \in \mathbb{C}$, kde $a, b \in \mathbb{R}$, plat

$$\begin{aligned} z + \bar{z} &= (a + bi) + (a - bi) = 2a = 2 \operatorname{Re}(z), \\ z - \bar{z} &= (a + bi) - (a - bi) = 2bi = 2i \operatorname{Im}(z). \end{aligned}$$

Teda

$$\operatorname{Re}(z) = \frac{1}{2}(z + \bar{z}), \quad \operatorname{Im}(z) = \frac{1}{2i}(z - \bar{z}).$$

(1)

$$\begin{aligned} \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2) &= \frac{1}{4}(\langle u + v, u + v \rangle - \langle u - v, u - v \rangle) \quad (\text{defincia normy}) \\ &= \frac{1}{4}(\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &\quad - \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle - \langle v, v \rangle) \quad (\text{linearita skalrneho sinu}) \\ &= \frac{1}{2}(\langle u, v \rangle + \langle v, u \rangle) \\ &= \frac{1}{2}(\langle u, v \rangle + \overline{\langle u, v \rangle}) \quad (\text{skorosymetria}) \\ &= \operatorname{Re}\langle u, v \rangle. \end{aligned}$$

(2)

$$\begin{aligned} \frac{1}{4}(\|u - iv\|^2 - \|u + iv\|^2) &= \frac{1}{4}(\langle u - iv, u - iv \rangle - \langle u + iv, u + iv \rangle) \quad (\text{defincia normy}) \\ &= \frac{1}{4}(\langle u, u \rangle - i\langle u, v \rangle + i\langle v, u \rangle + \langle v, v \rangle \\ &\quad - \langle u, u \rangle - i\langle u, v \rangle + i\langle v, u \rangle - \langle v, v \rangle) \quad (\text{linearita skalrneho sinu}) \\ &= \frac{1}{2}(i\langle v, u \rangle - i\langle u, v \rangle) \\ &= \frac{1}{2i}(\langle u, v \rangle - \overline{\langle u, v \rangle}) \quad (\text{skorosymetria}) \\ &= \operatorname{Im}\langle u, v \rangle. \end{aligned}$$